iii. Smelting (in a Blast Furnace): The charge consisting of roasted ore, coke and limestone in the ratio 8:4:1 is smelted in a blast furnace by introducing it through the hopper arrangement at the top. There are three important regions in the furnace.

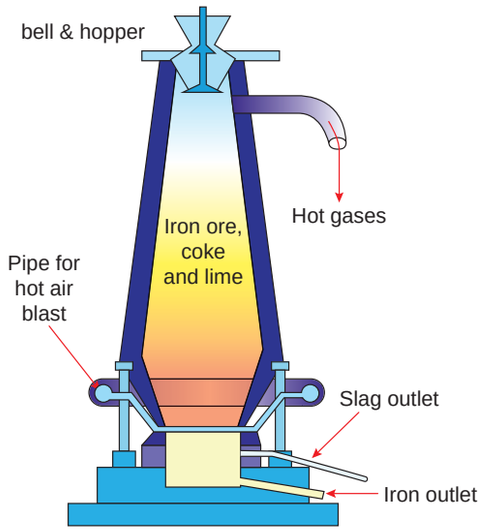


Figure 8.11 Blast Furnace

(a) The Lower Region (Combustion Zone)- The temperature is at 1500°C. In this region, coke burns with oxygen to form CO_2 when the charge comes in contact with a hot blast of air.

$$C + O_2 \xrightarrow[\Delta]{1500°C} CO_2 + \text{Heat}$$

It is an exothermic reaction since heat is liberated.

(b) The Middle Region (Fusion Zone) – The temperature prevails at 1000°C. In this region, CO_2 is reduced to CO.

$$CO_2 + C \xrightarrow[\Delta]{1000°C} 2\,CO - \text{Heat}$$

Limestone decomposes to calcium oxide and CO_2.

$$CaCO_3 \xrightarrow[\Delta]{1000°C} CaO + CO_2 - \text{Heat}$$

These two reactions are endothermic due to absorption of heat. Calcium oxide combines with silica to form calcium silicate slag.

$$CaO + SiO_2 \longrightarrow CaSiO_3$$

(c) The Upper Region (Reduction Zone)- The temperature prevails at 400°C . In this region carbon monoxide reduces ferric oxide to form a fairly pure spongy iron.

$$Fe_2O_3 + 3CO \xrightarrow{400°C} 2Fe + 3CO_2 \uparrow$$

The molten iron is collected at the bottom of the furnace after removing the slag.

The iron thus formed is called pig iron. It is remelted and cast into different moulds. This iron is called cast iron.

Physical properties:

i. It is a lustrous metal, greyish white in colour.
ii. It has high tensility, malleability and ductility.
iii. It can be magnetized.

Chemical properties:

i. **Reaction with air or oxygen:** Only on heating in air, iron forms magnetic oxide.

$$3\,Fe + 2\,O_2 \longrightarrow Fe_3O_4 \text{ (black)}$$

ii. **Reaction with moist air:** When iron is exposed to moist air, it forms a layer of brown hydrated ferric oxide on its surface. This compound is known as rust and the phenomenon of formation of rust is known as **rusting**.

$$4\,Fe + 3\,O_2 + x\,H_2O \longrightarrow 2\,Fe_2O_3 \cdot xH_2O \text{(rust)}$$

iii. **Reaction with steam:** When steam is passed over red hot iron, magnetic oxide is formed.

$$3\,Fe + 4\,H_2O\,(\text{steam}) \longrightarrow Fe_3O_4 + 4\,H_2 \uparrow$$

iv. **Reaction with chlorine:** Iron combines with chlorine to form ferric chloride.

$$2Fe + 3Cl_2 \longrightarrow 2FeCl_3 \text{ (ferric chloride)}$$

v. **Reaction with acids:** With dilute HCl and dilute H_2SO_4 it liberates H_2 gas.

$$Fe + 2HCl \longrightarrow FeCl_2 + H_2 \uparrow$$

$$Fe + H_2SO_4 \longrightarrow FeSO_4 + H_2 \uparrow$$

With dilute HNO_3 in cold condition it gives ferrous nitrate and ammonium nitrate.

$$4\,Fe + 10\,HNO_3 \rightarrow 4\,Fe(NO_3)_2 + NH_4NO_3 + 3\,H_2O$$

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